MINI PROJECT REPORT

on

HOSPITAL FINDER

Submitted in partial fulfilment for the completion of

B.E., IV Semester

in

INFORMATION TECHNOLOGY

By

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CERTIFICATE

This is to certify that the project work entitled "**HOSPITAL FINDER**" submitted to **CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY,** in partial fulfilment of the requirements for the completion of IV semester of B.E. in Information Technology, during the academic year 2019-2020, is a record of original work done by **MANAVI REDDY VEMULA (1601-18-737-010), SUMEDHAA MEDAVARAPU (1601-18-737-019)** during the period of study in Department of IT, CBIT, HYDERABAD, under my supervision and guidance.

Project Guide	**Head of the Department**
Dr.M.Venugopala Chari	**Dr.Suresh Pabboju**
Professor, Dept. of IT,	Professor, Dept.
CBIT, Hyderabad.	CBIT, Hyderabad.

DECLARATION

This is to certify that the work reported in the present report titled **“**Hospital Finder**”** is a record of work done by us in the Department of Information Technology, Chaitanya Bharathi Institute of Technology, Hyderabad.

No part of the report is copied from books / journals / internet and wherever the portion is taken, the same has been duly referred. The reported results are based on the project work done entirely by us and not copied from any other source.

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Sumedhaa Medavarapu (1601-18-737-019)

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ACKNOWLEDGEMENT

Before we get into details of the project, we would like to add a few words of appreciation for the people who have been a part of this project right from its inception.

It gives us immense pleasure in presenting this project report on **“**Hospital Finder **”**. The final outcome of this assignment required a lot of guidance and assistance from many people and we are extremely fortunate to have got this all along the completion. We would like to thank our principal **Dr.P.RAVINDER REDDY** and our head of the department **Dr.SURESH PABBOJU**. We would also like to thank **Dr.M.VENUGOPALA CHARI**, our project guide, for giving us an opportunity to do this project work and providing us all support and guidance to complete the project on time.

The cooperation among us and the efforts that we, **MANAVI REDDY VEMULA** and **SUMEDHAA MEDAVARAPU,** put in to complete this project helped us achieve this outcome. We would like to express our heartfelt gratitude to our friends and respondents for their support and their willingness to spend time with us.

ABSTRACT

The domain of this project comes under smart communication which involves designing devices which would help in easing communication channels between various communication devices and points and also comes under software-mobile app development.

Services and delays in calling for care, may result in loss of life, so the main motivation for doing this project is to facilitate users to search the nearest hospital based on their location.

The objective of this project is to create an app using Android Studios. The front end of the app will have user interface and visualization of data, where the user can interact with the app icons and can view the nearest hospital. The language used to develop the front end is Java. For storing data like the location of the hospitals and the users' information in the back end, Firebase database is used. The users' location is detected by using GPS, which is inbuilt in the device of the users.

LIST OF FIGURES

1. INTRODUCTION

1.1 OVERVIEW

WeCare is the name given to the Hospital Finder app, which is developed using Android Studios and Firebase database. This app is basically designed to help the user view the nearest hospital from their location. There are web applications based on this concept but as there is immense growth in technology and as people prefer mobile phones to desktops or laptops designing of mobile applications has come into the picture. Even though there are few mobile applications related to this concept, they have few drawbacks. To overcome these drawbacks WeCare app is designed.

This app consists of a home page, login page, registration page, and the page which shows the user's location and the hospital's location. If one is not the member/user of the app, then he/she should first register for the app by filling in the details. Once the user registers for the app, from then he/she can directly login in to the app. As soon as the user logs into the app, their location gets stored in the database. Initially latitudes and longitudes of the hospitals are also stored in the database. Now, the distance from the user's location to all the hospitals is calculated and the nearest hospital from the user's location is pinned on the map and distance in kilometers is also shown within no time.

1.2 MOTIVATION

The desire to create this project arose to let the user know the nearest hospital from their location which helps in taking the patient to the hospital on time, reducing the risk and panic factors.

The main objective of “Hospital Finder” app is to develop a mobile application that locates the nearest hospital from the users’ location. The nearest position of the hospital is calculated with a built-in feature i.e., Global Positioning System in smart phones and the distance from the users’ current location through Google map application(API).

Through this app, by locating the hospital nearer to the user, one can instruct the driver to take the patient to that particular hospital easily as services and delays in calling for care may result in loss of life.

1.3 PROBLEM STATEMENT

Many people face difficulties in identifying the hospitals nearer to them and end up taking the patient to the hospital which is far from them. This may lead to many unexpected situations. Though we have 108, unless we mention a specific hospital, it takes the victim only to the government hospital even if it is far. The ambulances which are associated with a particular hospital take the victim only to that hospital irrespective of the distance.

Therefore there is a need to introduce certain features so that the person will be able to guide the driver to the hospital that is closer to them resulting in less or no delay in calling for care.

2. EXISTING SYSTEM

In the existing system, when the user tries to find the nearest hospital, many hospitals are pinned on the map often confusing the user to decide the nearest hospital.

The present system may seek permission to access unnecessary data of the users like contacts and photos, by which privacy of the users may be lost and sometimes unless the user accepts the request they won't be directed to the next page. Therefore this may panic the user in an emergency.

There are few applications which show all the hospitals that are closer to the user only if he/she types "hospitals near me" which is quite a long process and sometimes it may not be accurate and may consume a lot of time.

Disadvantages of Existing System

- Lack of security and data accuracy.
- Maximum overhead when size of the database grows.
- Challenging task for maintaining the database.
- MySQL or other databases used in the existing system are not so flexible design-wise, new column insertion may affect design because they are mostly used for relational data and transactions.

3. PROPOSED SYSTEM

3.1 METHODOLOGY

To overcome the drawbacks of the existing system, this app is designed to locate one hospital which is nearest to the user. The application is so powerful that it can easily identify the user's location and then it can give the route to the nearest hospital through Global Positioning System(GPS) enabled navigation that precisely determines geographical location by receiving GPS coordinates information from the GPS satellites.

It also reduces manual work which includes searching for the nearest hospital and also helps in saving the user's time. One should just login in to the app and the app will identify the user's location and will calculate the distance from the user's location to all the hospitals and the closest hospital is pinned on the map. The distance is also displayed on the screen of the user i.e the distance from the user's location to the nearest hospital in kilometers. This app seeks permission to access the location of the user, only while using the app to identify the user's location and to help locate the hospital.

The back end used for this application is the Firebase database. When compared to the existing systems' databases, Firebase is suitable for real time application. In Firebase any Key/Field can be added easily without affecting the existing design. Firebase authentication gives backend services, simple to-use SDKs, and instant UI libraries to confirm clients over the application. It supports authentication using passwords, email id or username, therefore it is secure and the results are accurate.

3.2 ARCHITECTURE OF PROPOSED SYSTEM

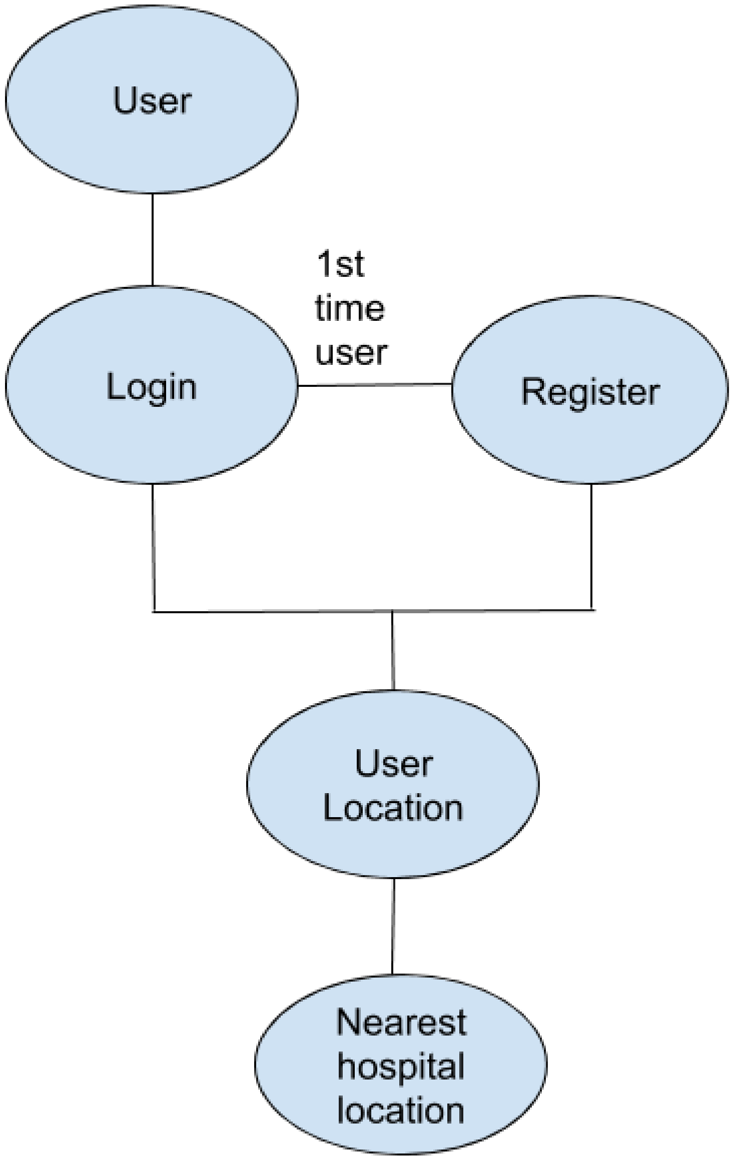

Fig 3.1

Architecture of the proposed system focuses on looking at the system as a combination of many different components, and how they interact with each other to produce the desired result. First the user has to login into the mobile application. If he\she is a first time user of the app, they have to get registered for the app by clicking on 'Don't have an account?' in the login page. Both the login and registration page directs the user to the same page i.e the user location and thus shows the nearest hospital to the user from their location.

4.SOFTWARE AND HARDWARE REQUIREMENTS

SOFTWARE REQUIREMENTS

- Operating Systems: Windows*7
- Android Studio (for front end)
- Java (programming language)
- Firebase Database (for back end)
- Application Program Interface 24 (API 24)
- Google Positioning System (GPS)

HARDWARE REQUIREMENTS

- Input Devices: keypad
- Sufficient RAM to run the app

5. IMPLEMENTATION OF PROJECT

5.1 IMPLEMENTATION

There are various functions which are included in the customer login section, customer registration section, and the map section. These functions enable us to perform different tasks like registering one's self for the app, and later logging in to the app, tracking the user's location and then finally showing the closest hospital.

The front end of the app is developed using Android Studios and the programming language used is Java. App's front end consists of the user interface and visualization of data, where the user can interact with the app icons. The back end of the app is developed using the Firebase database where the data of the users and hospitals is stored

There are three modules in this application namely login module, registration module and finding the nearest hospital based on the user location module. The users have to register by entering their username and password. This username is unique to all the users. This module helps to collect complete and relevant users' information.

The next module locates the nearest hospital. The nearest hospital is calculated with a built in feature of Global Positioning System in smartphones and displays the distance from the user's current location through google map Application Program Interface(API). The API key for Google Maps-based APIs is defined as a string resource. The API key used for this application is API 24 so that the application will be compatible on almost all the devices. Following is the code snippet of the application 'Hospital Finder'.

```
NearestHospitalButton.setOnClickListener(new View.OnClickListener() {
  @Override
  public void onClick(View v)
  {
     GeoFire geoFire = new GeoFire(CustomerRequestRef);
                              geoFire.setLocation(CustomerID,    new    GeoLocation(lastLocation.getLatitude(),
lastLocation.getLongitude()), new GeoFire.CompletionListener() {
        @Override
        public void onComplete(String key, DatabaseError error) {

        }
 CustomerLocation = new LatLng(lastLocation.getLatitude(),lastLocation.getLongitude());

mMap.addMarker(new MarkerOptions().position(CustomerLocation).title("You're Here.."));

NearestHospitalButton.setText("Searching nearest hospital...");
GetClosestHospital();
```

The functionality of the above code snippet is to locate the nearest hospital from the current location of the user. During the implementation of the project, in the backend i.e in the Firebase Database the longitudes and latitudes of all the hospitals are stored. This function runs when the user successfully logins from his current location.The user's coordinates are then stored in the database by 'Customer Location' and then displays the location of the user on the map with a pin showing 'You're Here'. The function then calculates the nearest hospital to the user displaying a message saying 'Searching nearest hospital…'.

5.2 RESULTS

Screenshot of Home Page

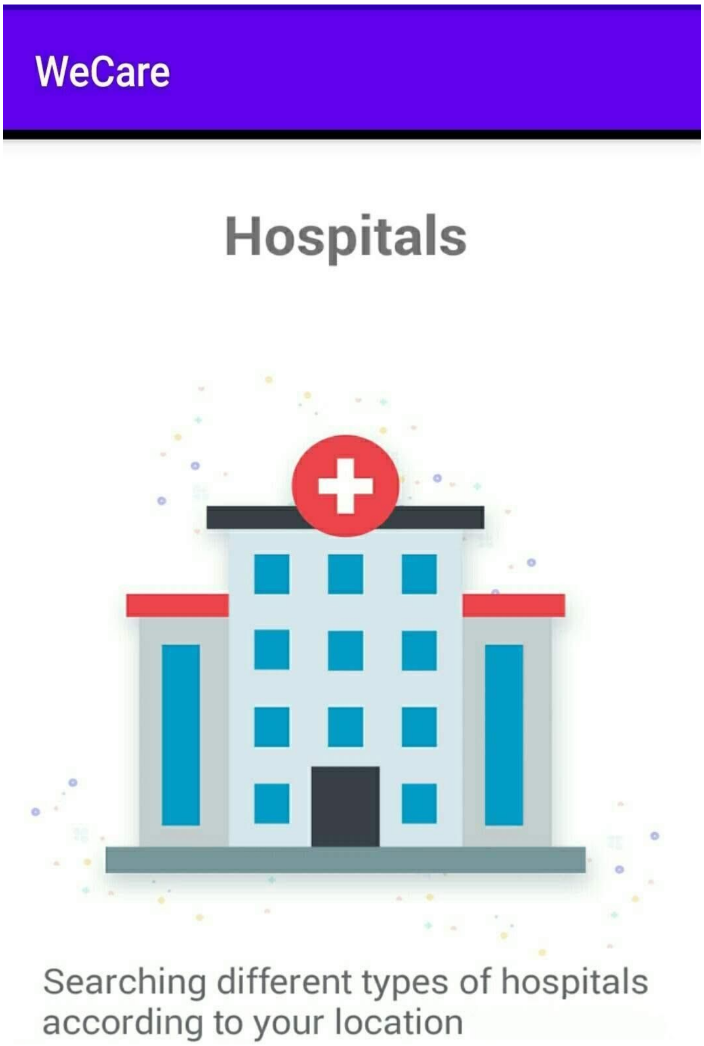

Fig 5.1

WeCare is the name given to the app. A picture is then inserted, which makes the home page of the app. It tells about the functioning of the app i.e., the basic plot of the app as shown in Fig 5.1.

<u>Screenshot of Login Page</u>

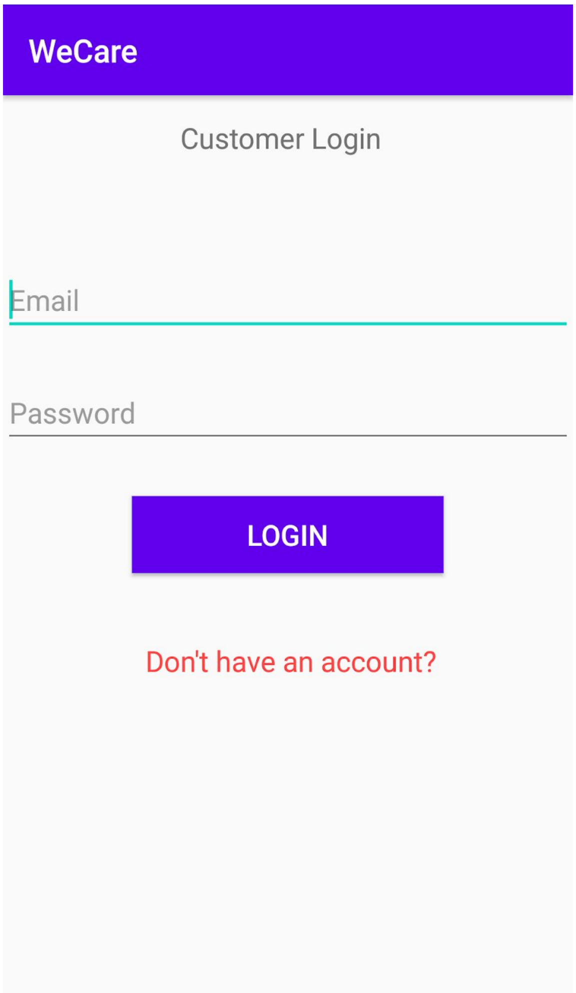

Fig 5.2

If the user already has an account, the login page appears to the user as shown in Fig 5.2. In case if the user is using the app for the first time they have to register as a user for the app by clicking on "Don't have an account?" If the user enters the wrong email or password a message saying "Incorrect email/password" appears.

Screenshot of Registration Page

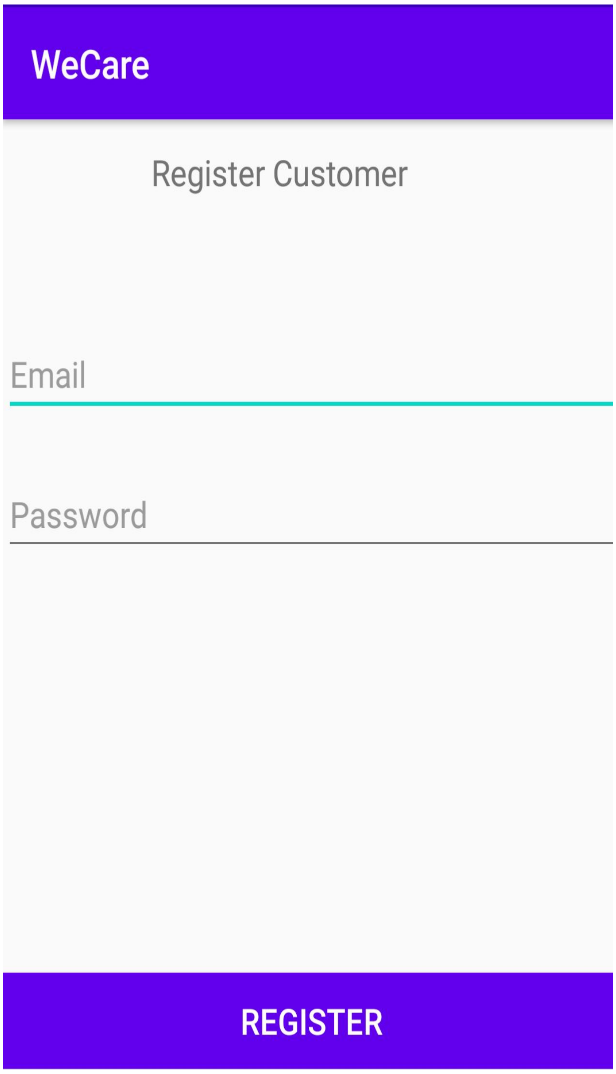

Fig 5.3

Through the registration page, the users can register by entering details like email and password as shown in Fig 5.3. This is necessary to know by whom all the app is being used. Every user gets a unique id which is stored in the database indicating who all have requested for viewing the nearest hospital.

Screenshot of User's Location

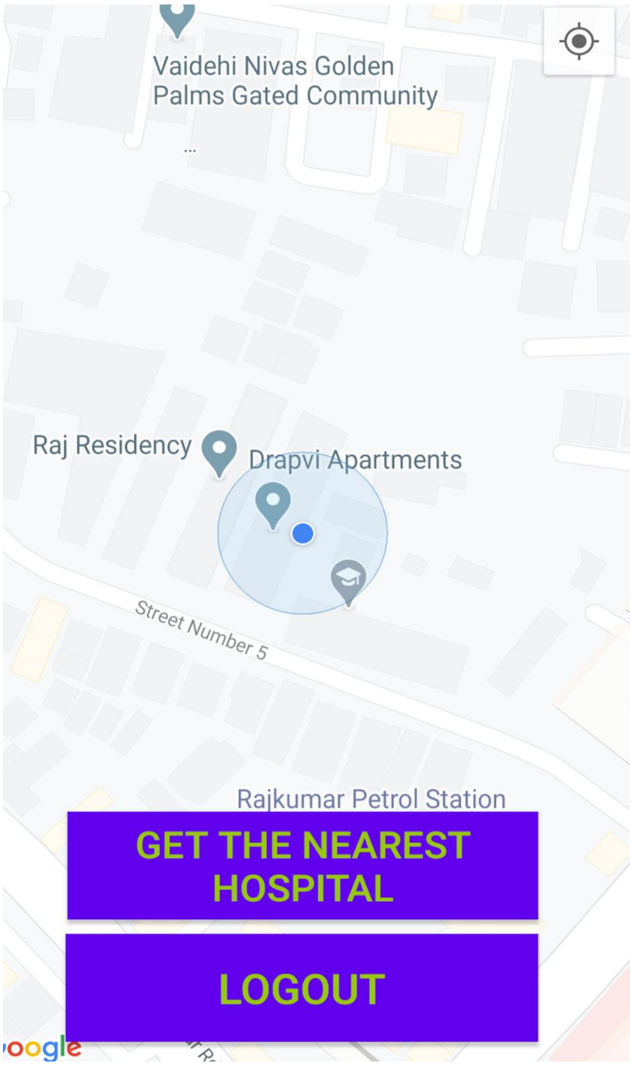

Fig 5.4

Once the user registers for the app the location of the user gets stored in the database and is pinned on the map as shown in Fig 5.4. The user's location is detected by the GPS, which is the software inbuilt in the users' device.

<u>Screenshot of Hospital's Location</u>

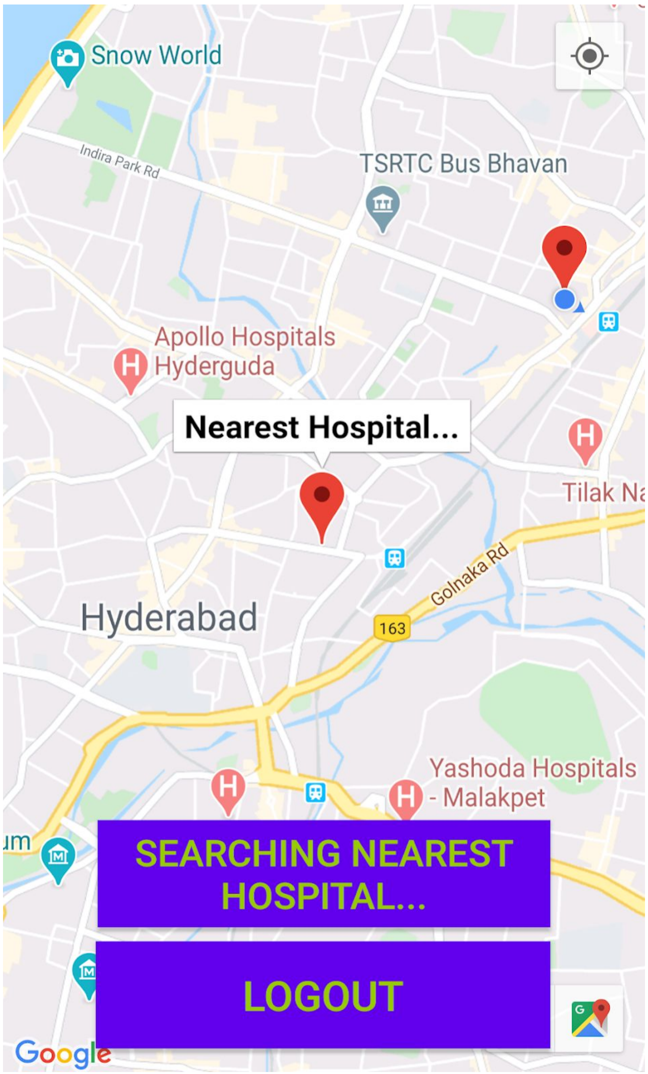

Fig 5.5

Initially, latitudes and longitudes of various hospitals are stored in the database. When the user clicks on "Get the nearest hospital" as shown in Fig 5.4, the distance from the users location to all the hospitals is calculated and the nearest hospital is pinned on the map as shown in Fig 5.5.

6. CONCLUSION AND FUTURE SCOPE

As stated before, the main objective of this project is to develop an user friendly app, to help people find the nearest hospital from their location. As the mobile application is easy to use and due to immense growth in technology, this application will have a lot of potential customers. The project has achieved what it set out to accomplish, leaving room for further improvement and enhancement.

The future scope of this project is to create such a software where the users will also have the opportunity to view the details of the hospitals like the contact number and hospital ambulance number. It can include other features like being able to view the specializations offered in the hospital and to call the hospital directly from the app.

7. BIBLIOGRAPHY

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